Supercooling – the energetics of freezing

Class practical

Students melt **sodium thiosulfate** crystals, and these are then cooled to a state well below the melting point. The melt now exists in a metastable **supercooled** state. The supercooled liquid will freeze rapidly on the addition of a crystal of sodium thiosulfate or dust particles, or on stirring. Students add a crystal of the solid and this seeds the **crystallisation** process. Temperature changes are observed throughout.

Lesson organisation

This is best carried out individually or in pairs. The experiment takes between 20 and 30 minutes.

Apparatus	Chemicals
Each group will need: **Test-tube (Note 1)** **Test-tube holder** **Stirring thermometer (-10 – 110°C) (Note 2)** **Beaker (100 cm^3)** **Bunsen burner** **Tripod and gauze** **Cotton wool, small tuft to fit test-tube**	Sodium thiosulfate-5-water, about 20 g **Refer to Health & Safety and Technical notes section below for additional information.**

Health & Safety and Technical notes

Read our standard health & safety guidance

Wear eye protection.

Sodium thiosulfate-5-water, $Na_2S_2O_3.5H_2O(s)$ - see CLEAPSS Hazcard.

1 The test-tube must be very clean.

2 A temperature sensor attached to a computer can be used in place of a thermometer. Thus a continuous plot can be drawn as the temperature changes occur.

Procedure

a Half fill a very clean test-tube with crystals of sodium thiosulfate-5-water.

b Put a tuft of cotton wool in the top of the test-tube to exclude dust.

c Warm the test-tube **gently** in a beaker of hot water (about 50 ^oC) to melt the crystals.

d When all the crystals have melted, remove the cotton wool, put a thermometer in the melt and record the temperature. If the

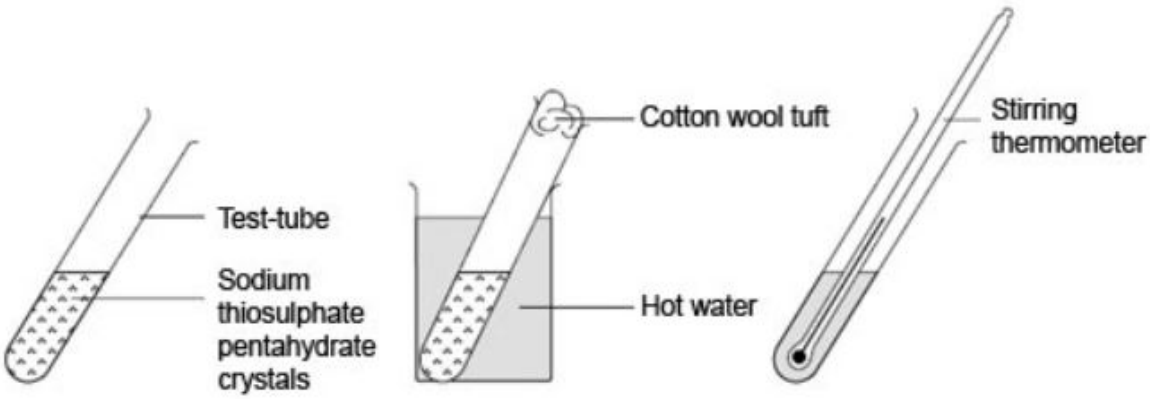
liquid starts to crystallise on inserting the thermometer, re-heat in water to melt all the solid.

e Stand the test-tube in an empty beaker and leave to cool where it won't be disturbed.

f Observe the temperature at various intervals until the value is in the region of 30 – 40 °C. No crystallisation should have occurred.

g Now add a fresh crystal of sodium thiosulfate, observe the rapid crystallisation which occurs, and once again continue to monitor the temperature at regular intervals.

h Wait until the temperature has fallen to about 25 – 30 °C.



Teaching notes

Once the experiment is over the thermometer can easily be removed by flushing with water, since sodium thiosulfate is water-soluble, or by re-melting the solid. Do not heat the test-tube directly over a Bunsen flame as at higher temperatures the thiosulfate decomposes and may form toxic products.

The temperature changes occurring show a steady fall as the liquid cools. When a crystal is added to the supercooled liquid, the temperature rapidly rises as solidification takes place, confirming this process is exothermic. The solid then cools to room temperature.

All solids exhibit supercooling to a greater or lesser extent, but sodium thiosulfate is particularly prone to exhibiting this metastable condition. With more time available, it is possible to cool down the melt to a value well below room temperature, but to achieve this involves waiting for more time to elapse, thus lengthening the experiment considerably.

An impressive teacher demonstration can be carried out using cold running water to cool the melt down rapidly to about 5 – 10 °C before a crystal is added to seed the crystallisation process. The test tube warms up extremely rapidly and becomes quite hot in the process.

Health & Safety checked, September 2014

Credits

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